A Multimodal Unified Representation Learning Framework With Masked Image Modeling for Remote Sensing Images

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***Abstract*— The coordinated utilization of diverse types of satellite sensors provides a more comprehensive view of the Earth's surface. However, due to the significant heterogeneity across modalities and the scarcity of high-quality labels, most existing methods face bottlenecks in the underutilization of massive unlabeled multimodal satellite data, making it challenging to understand the scene comprehensively. To this end, we propose a multimodal unified representation learning framework (MURLF) based on masked image modeling (MIM) for remote sensing (RS) images, aiming to make better use of massive unlabeled multimodal RS data. MURLF leverages the consistency and complementarity relationships among modalities to extract both common and distinctive features, mitigating the challenges faced by encoders due to significant heterogeneity across various data types. In addition, MURLF uses multilevel masking independently across different modalities, using visual tokens both within the same modality and across modalities to jointly recover masked pixels as the pretext task, facilitating comprehensive cross-modal information interaction. Furthermore, we design a preselected sensor-specific feature extractor (PSFE) to exploit the heterogeneous characteristics of various data sources, thereby extracting discriminative features. By integrating the multistage PSFE with the ViT backbone, MURLF can naturally extract multimodal hierarchical representations for downstream tasks, fully preserving valuable information from each modality. The proposed MURLF is not restricted to multimodal inputs but also supports single-modal inputs during the fine-tuning stage, significantly broadening the framework's application. Extensive experiments across multiple tasks demonstrate the superiority of the proposed MURLF compared with several advanced multimodal models. The code will be released soon.**

***Index Terms*— Cross-modality, masked image modeling (MIM), multimodal remote sensing (RS) data, pretraining, self-supervised learning (SSL).**

I. Introduction

THE utilization of various satellite sensors enables the acquisition of extensive multimodal remote sensing (RS) data, providing detailed and comprehensive information about the Earth's surface. This joint utilization supports diverse applications, including climate monitoring [1], [2], resource assessment [3], and land use classification [4], [5]. However, challenges such as complex observation scenarios, diverse imaging mechanisms, and varying data quality result in significant differences in data characteristics among modalities. In addition, the high cost of manual annotation in RS scenarios leads to most available images lacking labels, hindering efficient joint utilization of multimodal data.

Early research methods, such as multiple kernel learning (MKL) [6], [7], [8] and sparse representation [9], [10], attempted to exploit latent correlations between different modalities. With the advent of computational resources, CNN and transformer-based methods have shown promising results on limited datasets [11], [13], [14]. However, these supervised models heavily depend on high-quality labeled data and struggle to leverage the vast amount of unlabeled multimodal RS data, compromising their robustness and generalizability.

Self-supervised learning (SSL) approaches, such as instance discrimination (ID) and masked image modeling (MIM) [15], can extract generic representations from unlabeled samples. While ID aims to acquire discriminative instance representations and has been explored in various RS studies [16], [19]. However, it faces challenges with dense prediction tasks and often offers marginal performance improvements compared with supervised methods [20], [21], [22]. In contrast, MIM-based methods predict substantial masked information to learn enriched semantic representations, demonstrating success in both computer vision (CV) and RS field [23], [24], [25], [26], [27], [28], [29], [30], [31].

However, due to the significant heterogeneity across modalities and the scarcity of high-quality labels, there remains a lack of effective methods to fully leverage the vast amounts of unlabeled multimodal satellite data, thereby hindering comprehensive scene understanding. Specifically, most existing multimodality learning (MML) approaches rely heavily on labeled data, limiting their effectiveness with large unlabeled datasets. In addition, these multimodal fusion strategies depend on high-quality labels, making it difficult to adapt to self-supervised models. Although some methods have explored integrating MML with SSL [26], [27], [32], [33], [34], [35], they often fail to fully leverage deeper interactions between modalities and overlook the impact of sensor imaging mechanisms and the complementary relationships among modalities. Furthermore, while the ViT backbone has demonstrated computational efficiency with the MAE framework for mul-

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timodal images, it struggles to effectively use the multiscale information inherent in RS images. As a result, there remains a lack of MIM-based methods that can effectively exploit the vast amounts of unlabeled multimodal data to harness heterogeneous information.

To tackle these challenges, we present a multimodal unified representation learning framework (MURLF) with MIM for RS images to learn potential correlations among multimodal data. MURLF leverages the consistency and complementarity relationships between different modalities to extract both common and distinctive features, enhancing the model's capability to effectively use diverse modal data. In the encoder stage, we independently apply multilevel masking operations for each modality, using same-modal and cross-modal information to reconstruct masked patches, facilitating comprehensive interaction between modalities. Besides, we only input visible tokens into the encoder to maintain computational efficiency, similar to MAE [23]. In addition, we design a multistage PSFE integrated with the ViT backbone, enabling the model to extract hierarchical, modality-specific features, thereby improving performance in downstream tasks. In the decoder stage, given the significant differences among modalities, we use separate decoders for each modality to ensure consistent output features in the semantic space. Extensive experimental results corroborate the effectiveness of MURLF. The primary contributions are outlined as follows.

1) A generic MIM-based MURLF for RS images is proposed, which exploits the consistency and complementarity relationships among modalities to tackle the challenges posed by the significant differences in the characteristics of massive heterogeneous data inputs.
2) A preselected sensor-specific feature extractor (PSFE) is designed to leverage the heterogeneous characteristics of various data sources to extract discriminative features. By integrating the multistage PSFE with the ViT backbone, MURLF can naturally extract multimodal hierarchical representations for downstream tasks, thereby fully preserving the valuable information from each modality.
3) Same-modal and cross-modal visible tokens are used to jointly recover masked information as a pretext task, facilitating the establishment of connections among data from different modalities in the latent space.
4) The proposed MURLF accommodates both multimodal and single-modal inputs in downstream tasks, significantly expanding its applicability.

II. Related Work

In this section, we introduce two topics related to our study, including MML and SSL in RS.

A. MML in RS

MML is integral to various RS applications, with its core emphasis on the interaction of diverse modal data [36], [37], [38], [39]. Initially, researchers focused on information fusion through kernel learning. Zhang et al. [7] proposed an ensembled MKL framework combined with active learning, ensuring robust classification of multimodal data. Building on this, Gu et al. [6] introduced a novel MKL model using Gaussian kernels to enhance feature extraction. Further refinement came from Sun et al. [8], who merged features from optical and SAR images using MKL, achieving precise mapping of impervious surface distribution.

The sparse representation subsequently emerged as a prominent paradigm within MML. Rasti et al. [9] used extinction profiles to separately extract different modal features, followed by low-rank fusion based on sparse representation. Zhang and Prasad [10] used multitask joint sparsity modeling, validated with HSI-LiDAR data, while Dian and Li, [40] introduced a multirank regularization model to leverage correlations between LR-HSI and HR-MSI. Other methodologies, such as manifold learning [41], random forests [42], and tensor representation [43], have also demonstrated impressive performance in the joint utilization of multimodal data.

Recently, several CNN-based methods have been used for MML in RS [44], [45]. Kang et al. [46] developed a CNN-based model for land cover segmentation by analyzing the complementary properties of optical and SAR data. Zheng et al. [47] presented a framework integrating category-prior knowledge with enhancement learning for more effective information interaction. Hong et al. [11] unified five CNN- based fusion methods into a multimodal deep learning framework. In addition, transformer-based approaches have shown outstanding performance [48], [49], [50]. Fan et al. [51] proposed a transformer-based approach for urban informal settlements mapping. Yao et al. [52] extended Vision Transformers to multimodal domains, using cross-modality attention to enhance information interaction among heterogeneous data for land cover classification in RS.

However, most of the methods mentioned above rely on high-quality labeled datasets. Despite the abundance of multimodal RS data, only a tiny fraction is manually annotated. Consequently, these methods struggle to fully use the extensive multimodal data, resulting in models with limited generalization capabilities and robustness.

B. SSL in RS

SSL leverages the inherent characteristics of data to acquire generalized features applicable to various downstream tasks. The increasing availability of unlabeled data has made SSL combined with pretraining particularly appealing in the RS field [53]. SSL methods primarily include ID methods, such as MOCO [54], DINO [55], and Barlow Twins [56], and MIM methods, such as BEIT [57], MAE [23], and SimMIM [24]. ID methods typically enhance discriminative features by emphasizing similarities across different views of the same image and avoiding representational collapse [15]. Stojnić and Risojević [16] pioneered the application of CMC to RS, conducting extensive experiments to validate the efficacy of ID methods. Marsocci and Scardapane [58] used Barlow Twins in RS to extract robust features for segmentation, while Li et al. [18] integrated geographic coordinate information with RS imagery to provide additional constraints for representation learning. Wanyan et al. [19] combined global–local view alignment with DINO to demonstrate the effectiveness

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of multisized views in RS images. Despite their strengths, ID methods tend to excel in global decision-making tasks but face limitations in dense prediction tasks [20]. Recent studies indicated that ID methods often suffer from overfitting on pretrained datasets [59] and may yield limited performance improvements compared with supervised learning methods [21], [22].

Conversely, MIM methods, which focus on restoring masked pixels as a pretext task, have shown significant success in CV and are now being explored in RS [25], [26], [27], [28], [29], [30], [31]. Sun et al. [29] applied the SimMIM framework with Swin Transformer for RS, refining the pretext task specifically for dense prediction tasks. Cong et al. [30] introduced sequence information to better use spatial and temporal characteristics of images, while Wang et al. [31] proposed a rotated varied-size window attention to improve object representation acquisition during pretraining. Hong et al. [60] developed a 3-D generatively pretrained transformer to exploit spectral features in RS images, capturing both spatial and spectral characteristics. In addition, integrating ID and MIM methods has also yielded impressive performance in RS [61], [62], [63].

Despite these advancements, most existing SSL methods are designed for single-modal inputs. Practical applications, however, involve multisatellite coordinated Earth observation, providing data from multiple sensors simultaneously. Some efforts have extended SSL methods to accommodate multimodal inputs through early fusion [27], [32], [33], [34], but these approaches often fail to adequately consider data inherent characteristics or the complementarity among modalities. Consequently, models struggle to effectively use heterogeneous information. While Xue et al. [33] and Feng et al. [64] have demonstrated the efficacy of SSL in multimodal RS using limited datasets, there is still a scarcity of comprehensive research aimed at fully leveraging massive multimodal RS data to establish a unified multimodal representation learning framework based on MIM.

III. Methodology

The proposed MURLF, an MURLF with MIM for RS images, is depicted in Fig. 1. MURLF consists of multistage PSFEs, encoders, and decoders. This framework leverages the consistency and complementary relationships between different modalities to extract both common and distinctive features, effectively addressing challenges posed by the diverse data characteristics across modalities. Given that multimodal data input inevitably increases pre-training costs, we adopt an MAE-like architecture to enhance pretraining efficiency [23]. In this process, the ViT is selected as the backbone because it is more suitable for pretraining on large-scale datasets compared with CNNs [65]. Unlike most existing ViT-based methods that downsample feature maps directly to 1/16 of the original input size, MURLF can extract hierarchical sensor-related representations, thanks to the multistage PSFE design, which enables MURLF to demonstrate strong performance across various downstream tasks.

A. Masking Strategies

Extending self-supervised frameworks to accommodate multimodal inputs often involves the use of straightforward early fusion methods [27], [32]. However, such models typically prioritize the modality with higher information density, neglecting the other modality, particularly when there are substantial feature disparities between the different modal inputs. This could result in semantic confusion when distinct class targets exhibit similar characteristics within a single modality. In contrast, the proposed MURLF independently masks each modality, extracts features, and subsequently inputs them into the encoder. It uses both same-modal and cross-modal patches to collaboratively predict masked pixels as the pretext task. This approach facilitates extensive information interaction between different modalities, thereby reducing semantic confusion arising from imbalanced modal information.

Specifically, MURLF uses a multilevel masking strategy, as illustrated in Fig. 1(a) and (b), to mask feature maps at various scales. Initially, a binary mask vector of length L (Binary Mask 3) is generated using a random seed. This vector is reshaped into a matrix form and upsampled through interpolation to produce Binary Mask 2 $\in \mathbb{R}^{(H/8)\times(W/8)}$ and Binary Mask 1 $\in \mathbb{R}^{(H/4)\times(W/4)}$.

Binary Mask 3 is used to select masked tokens from the feature maps, ensuring that the encoder processes only the visible tokens, which enhances computational efficiency during pretraining. Binary Mask 1 and Binary Mask 2 are combined with multistage PSFE to mitigate the risk of information leakage [25], [66], as shown in Fig. 2. Consequently, during the pretraining phase, only the information from the visible patches is used. In the downstream tasks, the binary mask block is removed to ensure that the input information is complete.

B. Preselected Sensor-Specific Feature Extractor

In RS applications, images captured by different sensors can be categorized into two types based on the imaging mechanism: images with high interchannel correlation and images with low interchannel correlation. Commonly used optical sensors, such as visible, multispectral, and hyperspectral sensors, and the recorded energy values of ground features can be expressed as [37]

$$\mathrm{DN}_i = \int L(\lambda) R_i(\lambda) d\lambda \tag{1}$$

where $L(\lambda)$ represents the radiation energy at λ wavelength on the pupil of the lens. $R_i(\lambda)$ is the spectral response function, which measures the response of a sensor to different wavelengths of a continuous spectrum and satisfies $\int R_i(\lambda) d\lambda = 1$. The adjacent spectra display a high degree of correlation. As a result, there is a correlation between the energy values recorded in each channel and its closest neighbors, making it possible to effectively extract both the spectral and spatial features by simultaneously analyzing neighboring spaces and adjacent channels.

For SAR sensors, the intensity images obtained by different polarization modes measure the scattering characteristics in

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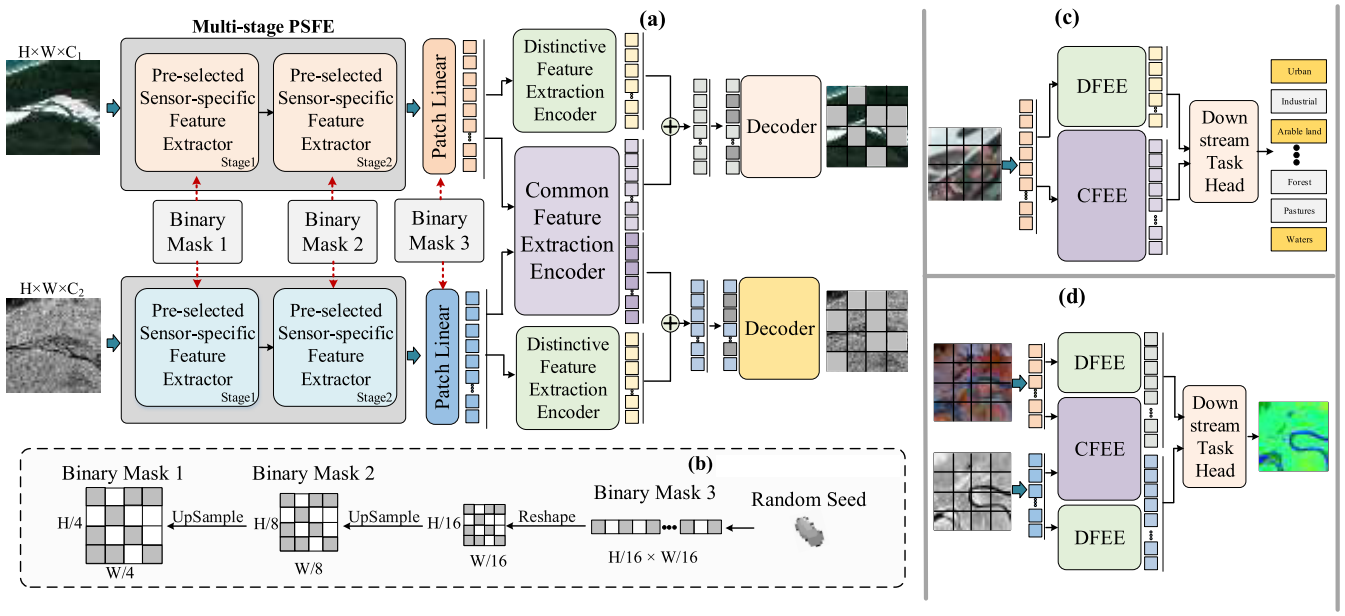


Fig. 1. Structure of MURLF. (a) Structure of MURLF for multimodal pretraining. The proposed MURLF consists of multiple multistage PSFEs, a CFEE for all modalities, multiple DFEEs for unique modalities, and decoders for multiple modalities. (b) Structure for multilevel masking, aimed at applying multiscale masks during the pretraining phase to ensure pretraining efficiency. (c) Structure of MURLF for single-modal fine-tuning. MURLF retains only the CFEE, the modality-specific DFEE, and the modality-specific multistage PSFE. (d) Structure of MURLF for multimodal fine-tuning. MURLF removes all decoders and connects the output features of the encoders directly to the downstream task head.

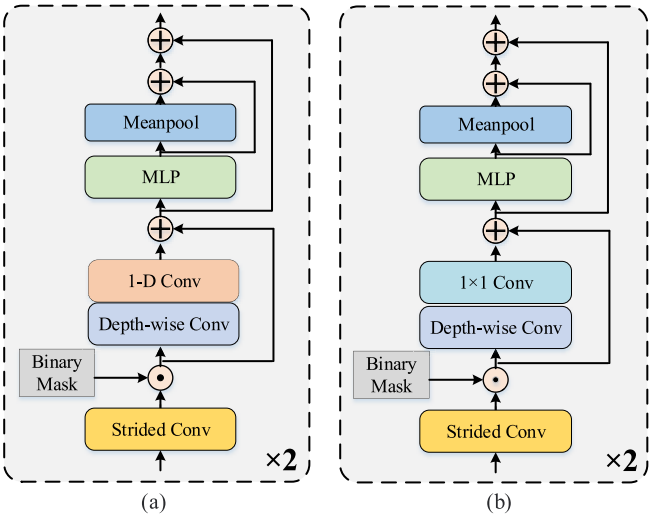


Fig. 2. Structure of multistage PSFE. (a) Multistage PSFE structure for images with high interchannel correlation. (b) Multistage PSFE structure for images with low interchannel correlation.

distinct directions. These characteristics are primarily influenced by the polarization ratio and relative amplitude, with low correlation between different channels. For infrared sensors, the data typically collected are single-channel images, thus requiring only the association of spatial information to extract distinctive features.

The designed PSFE possesses distinct structures for different types of inputs, enabling the comprehensive extraction of information inherent in each modality. The PSFE structure is similar to the transformer block but uses various convolution blocks to specifically capture the nearest-neighbor information from feature maps. After the PSFEs, the size of the output feature maps is $F_1 \in \mathbb{R}^{(H/4)\times(W/4)\times C_{F1}}$ and $F_2 \in \mathbb{R}^{(H/8)\times(W/8)\times C_{F2}}$, respectively. By integrating the multistage PSFE with the ViT backbone, MURLF can naturally extract multimodal hierarchical representations.

The details of the multistage PSFE are depicted in Fig. 2. PSFE initially adjusts the feature map size through a strided convolution, followed by extracting sensor-related features using sensor-specific attention. After the MLP operation, a mean pooling operation is added to introduce additional spatial interactions [67], expressed as follows:

$$x_{\text{attn}} = x_{\text{in}} \cdot \text{Mask} + \text{Attention}(x_{\text{in}} \cdot \text{Mask}) \tag{2}$$

$$x_{\text{out}} = x_{\text{attn}} + \text{MLP}(x_{\text{attn}}) + \text{Meanpool}(\text{MLP}(x_{\text{attn}})) \tag{3}$$

where x_{in} represents the downsampled feature maps obtained through a strided convolution, while Mask denotes the masked maps generated by the multilevel masking strategy. During the first stage of PSFE, the size of x_{in} is $(H/4) \times (W/4) \times C_{F1}$, and the size of the Mask is $(H/4) \times (W/4)$. In the second stage, the size of x_{in} is $(H/8) \times (W/8) \times C_{F2}$, and the size of the Mask is $(H/8) \times (W/8)$. MLP$(\cdot)$ refers to the multilayer perceptron operation, and Attention$(\cdot)$ signifies the sensor-specific attention operation, which fully accounts for the different characteristics of data acquired by various sensors. For input images with high interchannel correlation, we use depthwise convolution with a 5×5 kernel and 1-D convolution with a kernel length of 3 to extract nearest-neighbor spatial and spectral correlations, respectively, as shown in Fig. 2(a). This can be expressed as

$$\text{Attention}(\cdot) = \text{Conv}_{1\text{D}}(\text{Conv}_{\text{DW}}(\cdot)) \tag{4}$$

where $\text{Conv}_{\text{DW}}(\cdot)$ is the depthwise convolution while $\text{Conv}_{1\text{D}}(\cdot)$ represents the 1-D convolution.

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For input images with low interchannel correlation. The spatial correlation of the images is extracted using depthwise convolution and 1 × 1 convolution, which are depicted in Fig. 2(b)

$$\text{Attention}(\cdot) = \text{Conv}_{1\times1}(\text{Conv}_{\text{DW}}(\cdot)) \tag{5}$$

where $\text{Conv}_{1\times1}(\cdot)$ represents the 1 × 1 convolution.

C. Encoder

The encoder of MURLF consists of a common feature extraction encoder (CFEE) for all the modalities and distinctive feature extraction encoders (DFEEs) for each modality. This architecture facilitates learning a common representation across all the modalities and distinctive representations specific to each modality, respectively.

In this process, each transformer encoder layer incorporates a multihead self-attention along with a multilayer perceptron. Using multihead self-attention can help models understand complex and varied relationships within multimodal data, improving the capability of the model to capture correlations among features.

Given that different sensors capture various aspects of the same ground features, we maintain consistency by using identical positional embeddings across all the input modalities. This ensures that image regions described by tokens at the corresponding positions across modalities remain consistent. Following MAE, we use sine and cosine functions to generate these positional embeddings, expressed as [68]:

$$\text{PE}_{(\text{pos},2i)} = \sin\left(\frac{\text{pos}}{10\,000^{2i/d_{b\text{model}}}}\right) \tag{6}$$

$$\text{PE}_{(\text{pos},2i+1)} = \cos\left(\frac{\text{pos}}{10\,000^{2i/d_{b\text{model}}}}\right) \tag{7}$$

where $\text{PE}_{(\text{pos},i)}$ represents the value at the ith dimension of the posth position in the positional embeddings, and d_{model} denotes the total number of dimensions.

For different input modalities $x_{M1} \in \mathbb{R}^{H\times W\times C_1}$ and $x_{M2} \in \mathbb{R}^{H\times W\times C_2}$, tokens $\text{TK}_{M1} \in \mathbb{R}^{N\times D}$ and $\text{TK}_{M2} \in \mathbb{R}^{N\times D}$ are obtained after patch linear respectively, where N represents the number of tokens generated from each modality, and D represents the feature dimensions of each token. In the DFEE branch, the tokens are independently shuffled within each modality. In the CFEE branch, the tokens from two modalities are stacked together to form $\text{TK}_S \in \mathbb{R}^{2N\times D}$, followed by a shuffle operation on TK_S to extract robust features. Subsequently, the unshuffle operation returns these tokens to their original order. Finally, the common and distinctive features can be aggregated using the addition operation, resulting in a more comprehensive depiction of the ground features. Fig. 3 illustrates the positional changes in tokens from different modalities in the CFEE and DFEE branches.

D. Decoder

Considering the substantial differences among modalities, we use separate decoders for each modality to improve feature learning. The decoder structure is similar to that of the encoder, consisting of multiple cascaded transformer blocks. In the first layer of the decoder, we replace the self-attention block with a cross-attention block for effective information interaction. The motivation behind this step is to introduce information from tokens of other modalities via cross-attention [26], enabling the reconstruction of pixels corresponding to masked regions by jointly using both single- and cross-modal information. The cross-attention can be formulated as follows:

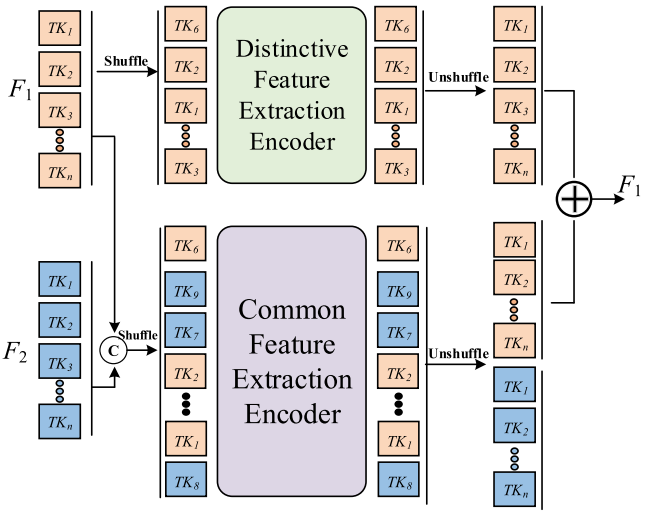


Fig. 3. Illustration of the positional changes in tokens.

$$\text{CrossAT}(Q_M, K_{\text{Vis}}, V_{\text{Vis}}) = \text{softmax}\left(\frac{Q_M \cdot K_{\text{Vis}}^{\text{T}}}{\sqrt{h_d}}\right) \cdot V_{\text{Vis}} \tag{8}$$

where $\text{CrossAT}(\cdot)$ represents the cross-attention mechanism, which explicitly establishes the interrelationships between the current modality and the remaining modalities. $Q_M \in \mathbb{R}^{N\times D}$ denotes the query vector of the current modality, while K_{Vis} and $V_{\text{Vis}} \in \mathbb{R}^{M\times D}$ refer to the key and value vectors, respectively, which are obtained by concatenating the visible tokens from the encoders of different modalities. h_d denotes the embedding dimension allocated to each head in the multihead attention mechanism. For the remaining transformer blocks, the self-attention is retained. In the final layer of the decoder, a reshape operation is applied to ensure that the sizes of the output features are consistent with the input images.

Similar to MAE [23], we use the mean squared error (MSE) loss to gauge the discrepancy between the reconstructed and actual images, restricting the loss computation to the masked patches, which can be expressed as

$$\begin{aligned}\mathcal{L} &= \mathcal{L}_{M1} + \mathcal{L}_{M2} \\ &= \frac{1}{m}\sum_{i\in M_1}^{m}\left(x_{M1i} - \hat{x}_{M1i}\right)^2 + \frac{1}{n}\sum_{j\in M_2}^{n}\left(x_{M2j} - \hat{x}_{M2j}\right)^2\end{aligned} \tag{9}$$

where m and n represent the quantities of masked patches from modalities M_1 and M_2, respectively, and x represents the patches from the reconstructed images, while $\hat{x}$ represents the patches from the actual images.

E. Structure for Fine-Tuning

As shown in Fig. 1, when the downstream task involves single-modal inputs, MURLF retains only the CFEE structure and the DFEE structure corresponding to the modality of

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the downstream task. The remaining structures are removed through network pruning. Specifically, we construct a new model that includes only the retained branches and transfer the corresponding weights from the original model. This approach effectively reduces the number of network parameters while maintaining high accuracy.

When the downstream task involves multimodal inputs, MURLF removes the decoders and performs a straightforward information fusion operation on the features from the various modalities.

Fig. 4(a) shows the MURLF structure for the classification task. MURLF concatenates the features output from the two modalities and subsequently inputs them into the linear classifier for fine-tuning. During the fine-tuning process, the class token is removed from the transformer blocks, and the global average pooling operation is applied to facilitate the utilization of heterogeneous features. This process is detailed as follows:

$$\text{Output} = \text{Linear}(\text{Concat}(F_{M1}, F_{M2})) \tag{10}$$

where F_{M1} and $F_{M2} \in \mathbb{R}^{N\times D}$ denote the features from modalities M_1 and M_2, respectively. N represents the number of tokens output by each modality, and D represents the dimensionality of each token.

Extracting multiscale features for semantic segmentation has been demonstrated to be effective in RS [69]. However, prior ViT-based approaches typically segment input images into multiple 16×16 patches and directly downsample them to 1/16 of the original size, thereby limiting the utilization of multiscale features. To address this limitation, the proposed MURLF extracts multiscale features at various stages and combines them through concatenation. This method enables the model to capture both global information and local details of the input image, enhancing semantic segmentation performance.

Fig. 4(b) displays the details of MURLF for the segmentation task. MURLF conducts concatenation operations on the output features of different modalities at the same scale separately, which includes features F_1 with the size of $(H/4) \times (W/4) \times C_{F1}$ from the first stage PSFE of different modalities, features F_2 with the size of $(H/8) \times (W/8) \times C_{F2}$ from the second stage PSFE of different modalities, features F_3 with the size of $(H/16) \times (W/16) \times C_{F3}$ from the encoders, and after the downsampling operation, features F_4 with the size of $(H/32) \times (W/32) \times C_{F4}$ are produced, which is achieved through a strided convolution. Followed by feeding the multiscale features into the UperNet head for fine-tuning [70], which can be represented as

$$\text{Output} = \text{UperNet}(F_1, F_2, F_3, F_4). \tag{11}$$

F. Implementation Details

The incorporation of multiple input modalities unavoidably increases the number of tokens. Considering the transformer blocks have quadratic computational complexity [65], the proposed MURLF uses a random masking strategy, inputting only the visible tokens into the encoder during the pretraining stage, thereby upholding accuracy while minimizing computational complexity [23].

We select two different parametric quantities of backbones, ViT-S and ViT-B [65], to validate the effectiveness of the proposed MURLF. For the ViT-S backbone, we set the CFEE depth to 12 and the DFEE depth to 2. For the ViT-B backbone, we set the CFEE depth to 12 and the DFEE depth to 4. Moreover, considering the high dimensionality and complexity inherent in RS data, we enhance the decoder's capacity by setting its depth to 8 for all the backbones.

IV. Experiments on Paired Optical–SAR Data

Due to the significant dissimilarity and complementarity between optical and SAR imagery [71], [72], along with the abundance of spatially aligned optical and SAR datasets, we initially opt to use paired optical–SAR data for conducting extensive experiments to validate the efficacy of the proposed framework.

A. Pretraining Setup

1) Dataset Description: We pretrain MURLF and other SOTA SSL-based approaches on the SSL4EO-S12 dataset[1] [32], which is a massive dataset consisting of one million pairs of optical–SAR images without labels. The optical data, consisting of 13 multispectral bands, are sourced from the Sentinel-2 satellite, while the SAR data, featuring both VH and VV polarizations, are acquired from the Sentinel-1 satellite. We use all the images within the SSL4EO-S12 dataset for pretraining and adjust their size to 224×224 pixels to meet the network's input specifications.

2) Experimental Settings: We conduct pretraining with the proposed MURLF for 100 epochs. The AdamW optimizer and cosine learning rate schedule are used for model optimization. The learning rate is set at $5e^{-4}$, while the batch size is specified as 256. In addition, a weight decay of 0.02 is incorporated. During pretraining, we use RandomResizedCrop, RandomHorizontalFlip, and RandomVerticalFlip for data augmentation.

We pretrain several SSL-based methods for equitable comparison on the SSL4EO-S12 dataset using the same settings as MURLF for 100 epochs, including DINO [55], MoCov3 [54], MAE [23], S2FL [33], and MultiMAE [26]. All the SSL-based methods adopt ViT-S as the backbone unless otherwise specified. Besides, following similar practices from previous studies [27], [32], [33], [34], we input cross-modal data into DINO, MoCov3, and MAE through early fusion for pretraining. In addition, we incorporate RandomSensorDrop [34] to ensure the validity of optical and SAR data simultaneously. All the experiments are implemented on four NVIDIA Tesla A100 GPUs during the pretraining stage.

B. Scene Classification

1) Dataset Description: Two classic multimodal scene classification datasets are used to evaluate the capability of MURLF.

So2Sat-LCZ42[2] is a comprehensive single-label dataset with paired optical–SAR images for local climate zone prediction [73]. The dataset consists of 400 673 image pairs and is

[1] https://github.com/zhu-xlab/SSL4EO-S12
[2] https://github.com/zhu-xlab/So2Sat-LCZ42

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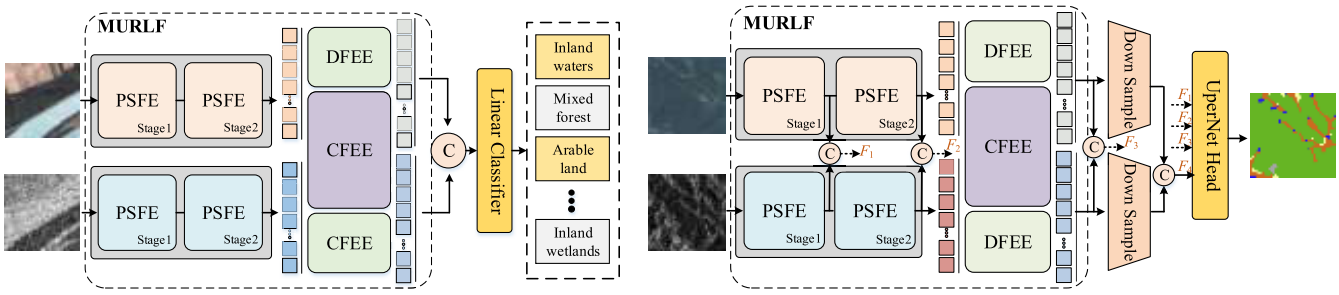


Fig. 4. Structure of MURLF for fine-tuning on multimodal RS interpretation tasks. (a) MURLF structure for classification task. (b) MURLF structure for segmentation task.

categorized into 17 distinct classes. Consistent with the official setup, we choose 352 366 pairs for training and 24 119 for testing. The training and test data are sourced from different cities, exhibiting significant differences.

BigEarthNet-MM[3] is a large-scale multilabeled dataset for multimodal RS scene classification, consisting of paired optical–SAR images [74]. The dataset comprises 590 326 pairs of images across 19 classes. We interpolate the spatial resolution of all the images to 10 m. Following the setting of [32] and [74], we allocate 311 667 pairs for training and 103 944 for testing.

2) Experimental Settings and Evaluation Metrics: A linear classifier is used as the task-specific header in the scene classification task. The learning rate is configured to 0.1, while the batch size is specified as 256. We choose the SGD optimizer together with a cosine learning rate schedule to optimize the model for 100 epochs. During the fine-tuning process for the scene classification task, the data augmentation methods used are RandomResizedCrop, RandomHorizontalFlip, and RandomVerticalFlip.

For the single-label scene classification task on the So2Sat-LCZ42 dataset, CrossEntropyLoss is used to guide the learning process, while overall accuracy (OA), average accuracy (AA), and accuracy in each class are used as evaluation metrics.

For the multilabel scene classification task on the BigEarthNet-MM dataset, MultiLabelSoftMarginLoss is used to guide the learning process, while $\text{mAP}_{\text{micro}}$, $\text{mAP}_{\text{macro}}$, and $\text{mAP}_{\text{sample}}$ are used as evaluation metrics, where $\text{mAP}_{\text{micro}}$ measures the overall performance of the model across all classes; $\text{mAP}_{\text{macro}}$ computes the average performance within each class; and $\text{mAP}_{\text{sample}}$ calculates the metrics for each instance, which is particularly suitable for measuring model performance in multilabel classification tasks.

3) Results and Analysis: The proposed MURLF and the compared SOTA methods for the multimodal RS scene classification task are shown in Tables I and II, respectively, where ResNet50 [75], ViT [65], ExViT [52], SCT [13], and Swin [76], are SL-based methods, while DINO, MoCov3, MAE, S2FL, and MultiMAE, are SSL-based methods. MURLF (ViT-B) represents using ViT-B as the backbone for MURLF, enabling a more extensive evaluation of the proposed MURLF's performance across a broader range of parameters.

As demonstrated in Table I, SSL-based approaches outperform SL-based approaches on average across the So2Sat dataset. Compared with randomly initialized ViT methods, which achieve an OA of 63.0% and an AA of 42.1%, all the SSL-based methods exhibit superior performance. This observation demonstrates the efficacy and dependability of the pretraining strategy on a sizable unlabeled dataset, followed by fine-tuning on a downstream task dataset. Notably, among the SSL-based methods, MultiMAE and the proposed MURLF significantly outperform others, consistent with expectations that deeper level feature interaction in multimodal data processing typically yields superior results compared with early fusion. Deeper feature interaction can better respect the unique characteristics of different modalities, thereby reducing semantic confusion. The proposed MURLF attains the highest accuracy on the So2Sat dataset and demonstrates optimal performance across several classes. Compared with the second-best method, the proposed MURLF improves OA by 2.5% and AA by 10.7%, proving its effectiveness. Furthermore, compared with the MURLF, the MURLF (ViT-B) exhibits enhanced performance with increased parameter counts, showcasing the scalability of the proposed method.

Table II provides further evidence of the effectiveness of MURLF for multilabel scene classification. MURLF demonstrates superior performance across all the evaluated metrics, showing improvements of 1.4% in $\text{mAP}_{\text{micro}}$, 2.6% in $\text{mAP}_{\text{macro}}$, and 1.0% in $\text{mAP}_{\text{sample}}$. In addition, the proposed method attains optimal results across all the classes, with the most significant advantage observed in the C17 (coastal wetlands) class, which encompasses diverse and complex features comprising aquatic and terrestrial elements. Therefore, the models have to combine characteristics from both optical and SAR modalities for accurate assessments. Compared with other methods, MURLF can efficiently use heterogeneous information based on the consistency and complementarity relationship between different modal data and learn multiscale discriminative visual representations using the characteristics of different data sources, which allows the model to develop a profound understanding of the scene, leading to the best performance in scene classification.

[3]https://bigearth.net/

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TABLE I
COMPARATIVE RESULTS OF DIFFERENT METHODS ON THE SO2SAT-LCZ42 DATASET FOR SCENE CLASSIFICATION

	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17	OA	AA
ResNet50	17.0	53.3	61.2	55.6	17.7	38.1	6.02	82.2	44.7	17.7	93.6	12.1	16.6	41.5	32.9	34.5	98.7	58.65	42.55
ViT	11.7	56.4	62.0	66.9	23.1	41.3	11.7	78.6	58.9	21.3	79.8	8.6	**31.1**	47.2	7.0	16.9	92.9	63.01	42.08
ExViT	17.3	**64.2**	55.8	59.8	21.3	40.4	11.0	80.7	60.9	24.3	90.4	7.9	12.1	50.2	45.1	25.6	98.9	62.42	45.05
SCT	15.9	55.8	56.4	53.2	22.4	42.3	19.2	79.4	62.8	27.6	94.8	12.1	24.2	56.3	40.2	28.7	**99.8**	63.69	46.53
Swin	19.1	59.1	**63.6**	56.1	20.7	42.8	8.6	74.3	61.8	27.2	94.5	12.8	21.8	54.2	27.6	34.7	96.1	65.16	45.58
DINO	30.4	60.3	55.5	62.4	22.0	38.5	17.4	80.8	55.2	32.5	92.9	15.5	22.4	54.2	33.2	32.5	99.2	64.20	47.34
MoCov3	27.8	52.7	57.0	62.2	23.5	**45.5**	15.0	79.0	64.1	19.2	92.8	10.6	37.1	52.1	11.6	33.7	99.6	65.19	46.08
MAE	12.9	55.8	60.4	47.9	21.1	40.4	19.7	86.8	63.4	17.5	92.9	18.5	14.9	**60.5**	48.9	34.6	95.2	65.62	46.55
S2FL	21.4	57.1	62.4	54.4	21.2	37.4	17.2	75.2	56.7	30.7	94.4	12.8	21.6	58.8	49.9	32.1	97.8	63.90	46.91
MultiMAE	24.5	54.8	58.8	55.1	19.7	42.8	14.0	86.1	68.2	26.5	90.4	19.8	22.8	56.3	30.6	27.2	**99.8**	66.22	47.57
MURLF	**34.3**	62.4	61.9	**76.0**	**50.2**	**45.5**	**25.2**	**89.2**	**80.6**	**53.6**	**97.6**	**40.1**	19.3	60.4	**51.9**	**42.8**	99.7	**68.72**	**58.28**
MURLF (ViT-B)	37.2	69.6	74.1	68.2	48.4	47.2	20.7	91.9	83.0	45.1	96.0	50.4	14.5	77.2	72.0	42.8	99.5	70.11	61.05

C1: Compact high-rise. C2: Compact mid-rise. C3: Compact low-rise. C4: Open high-rise. C5: Open mid-rise. C6: Open low-rise.C7: Lightweight low-rise. C8: Large low-rise. C9: Sparsely built. C10: Heavy industry. C11: Dense trees. C12: Scattered tree. C13: Bush, scrub. C14: Low plants. C15: Bare rock or paved. C16: Bare soil or sand. C17: Water.

TABLE II
COMPARATIVE RESULTS OF DIFFERENT METHODS ON THE BIGEARTHNET-MM DATASET FOR SCENE CLASSIFICATION

Method	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17	C18	C19	mAP_{micro}	mAP_{macro}	mAP_{sample}
ResNet50	85.1	55.3	91.2	70.5	85.3	77.5	74.5	86.4	87.1	94.6	91.4	64.7	75.2	75.7	72.5	71.0	58.8	92.2	98.3	88.20	79.33	90.95
ViT	83.1	53.3	92.3	75.1	86.9	75.7	73.6	85.5	87.4	94.3	90.2	67.5	77.2	75.5	70.1	70.5	58.7	91.1	97.6	88.16	79.24	90.60
ExViT	84.9	54.6	93.4	73.3	86.1	76.8	74.8	87.1	87.0	95.4	91.1	64.3	73.8	74.9	62.4	71.3	54.5	92.1	99.5	88.69	78.83	91.32
SCT	85.7	55.6	94.1	77.6	87.9	79.9	76.4	89.2	88.3	95.9	92.5	70.0	81.2	77.6	73.4	74.9	61.1	92.3	99.8	89.99	81.77	92.55
Swin	87.1	59.1	94.4	78.3	88.0	79.8	77.3	89.9	88.4	96.0	92.5	70.1	80.5	78.2	75.1	76.4	67.2	93.7	99.8	89.69	82.73	92.68
DINO	87.4	61.5	94.5	78.9	88.2	79.8	76.9	90.1	88.6	96.0	92.3	71.0	80.1	78.1	76.2	76.1	67.3	94.0	99.8	89.89	83.11	92.64
MoCov3	86.7	60.4	94.3	78.1	87.8	79.3	76.8	89.8	88.1	96.0	92.3	69.4	79.9	77.9	75.4	76.6	65.5	93.8	99.8	89.50	82.50	92.47
MAE	84.4	54.3	93.6	75.3	87.2	78.2	75.3	87.5	87.9	95.7	91.7	68.9	78.3	76.3	73.8	73.6	62.8	92.0	99.6	88.86	80.87	91.95
S2FL	86.4	59.1	94.3	76.5	87.2	79.2	75.7	89.2	88.2	95.8	92.2	67.2	79.0	77.4	73.5	74.8	62.9	93.6	99.8	89.23	81.67	92.31
MultiMAE	88.6	61.9	94.9	81.0	88.4	80.8	77.9	91.8	89.4	96.2	92.9	70.9	82.1	79.2	77.2	77.7	66.5	94.1	99.8	90.26	83.77	92.97
MURLF	**90.0**	**66.4**	**95.4**	**85.4**	**89.4**	**82.8**	**79.9**	**94.0**	**90.4**	**96.6**	**93.5**	**75.9**	**86.3**	**81.1**	**83.7**	**80.2**	**74.2**	**95.1**	**99.9**	**91.61**	**86.33**	**93.97**
MURLF (ViT-B)	91.3	69.6	96.1	87.5	91.5	85.6	82.1	94.7	91.7	97.1	94.5	80.5	89.4	83.6	83.9	84.5	79.4	95.7	99.9	92.37	88.33	94.73

C1: Urban fabric. C2: Industrial or commercial units. C3: Arable land. C4: Permanent crops. C5: Pastures. C6: Complex cultivation patterns. C7: natural vegetation. C8: Agro-forestry areas. C9: Broad-leaved forest. C10: Coniferous forest. C11: Mixed forest. C12: Natural grassland. C13: Moors, heathland and sclerophyllous vegetation. C14: Transitional woodland/shrub. C15: Beaches, dunes, sands. C16: Inland wetlands. C17: Coastal wetlands. C18: Inland waters. C19: Marine waters.

In practical applications, the amount of data with labels may be limited. Therefore, we evaluate the applicability of the various methods on the BigEarthNet-MM dataset with varying label sizes. Specifically, we select 1%, 5%, and 10% samples of the training set for scene classification, respectively. The results are depicted in Table III, which indicates that MURLF consistently achieves optimal performance across all the three dataset size settings.

C. Land Cover Segmentation

1) Dataset Description: Two classic multimodal land cover segmentation datasets are used to assess the ability of MURLF.

The DFC2020[4] is a widely used dataset for multimodal RS image segmentation [77], which contains 5128 optical–SAR pairs with high-resolution labels across eight classes to offer a thorough insight into land use patterns. The dimensions of the images are 264 × 264 pixels. We randomly select 3846 pairs for training and 1282 pairs for testing.

The WHU-OPT-SAR[5] is a publicly accessible optical–SAR paired dataset constructed for land cover use segmentation [78], encompassing seven ground feature classes.

[4]https://github.com/lukasliebel/dfc2020_baseline
[5]https://github.com/AmberHen/WHU-OPT-SAR-dataset

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TABLE III
COMPARATIVE RESULTS OF DIFFERENT METHODS ON THE BIGEARTHNET-MM DATASET WITH VARYING LABEL SIZES IN mAP_{sample} METRICS

	ResNet50	ViT	ExViT	SCT	Swin	DINO	MoCov3	MAE	S2FL	MultiMAE	MURLF
1%	78.98	80.38	79.37	80.68	82.80	83.91	83.28	83.45	83.68	84.27	**86.83**
5%	85.73	84.78	85.33	86.89	86.86	87.75	88.44	86.47	86.96	88.55	**89.91**
10%	87.05	87.87	87.85	88.86	89.01	89.85	89.18	88.73	88.59	89.93	**91.42**

The optical images are sourced from the GF-1 satellite and comprise four channels, while the SAR images are acquired from the GF-3 satellite and consist of one channel. Notably, the dataset exhibits significant differences from the SSL4EO-S12 dataset used for pretraining, enabling an evaluation of the model's fine-tuning performance across diverse data types. Following the setting of [78], we select 17 640 pairs for training and 5880 pairs for testing.

2) Experimental Settings and Evaluation Metrics: For the land cover segmentation task, we adopt UperNet as the segmentation head and use CrossEntropyLoss to guide the learning process. The learning rate is set to $6e^{-4}$, and the batch size is configured to 16. We use the AdamW optimizer for model optimization over 160k iterations. Besides, we use OA, mFscore, and mIOU as evaluation metrics to comprehensively assess the effectiveness of different methods. For the downstream segmentation task, the data augmentation methods used include RandomResize, RandomCrop, and RandomFlip, all of which are implemented using the mmsegmentation library.

3) Results and Analysis: Tables IV and V present the segmentation results of MURLF and other SOTA methods on the DFC2020 and WHU-OPT-SAR datasets, respectively. In these tables, Deeplabv3+ [79], MCANet [78], CMX [80], Tokenfusion [81], ViT + UperNet, and Swin + UperNet, are SL-based models, while DINO, MoCov3, MAE, S2FL, and MultiMAE, are SSL-based models.

We observe that the improvement of SSL-based models is limited compared with the advanced multimodal SL-based segmentation models, particularly on the WHU-OPT-SAR dataset. This may be attributed to the substantial difference in data distribution between the pretraining and fine-tuning datasets, which could adversely impact the performance of SSL-based models. Nonetheless, it is evident that MURLF consistently achieves SOTA performance even when merely connected to UperNet for segmentation tasks without additional complex multimodal feature interactions. It demonstrates a 1.5% increase in mIOU on the DFC2020 dataset and a 2.4% increase on the WHU-OPT-SAR dataset compared with the second-best method.

Fig. 5 demonstrates the segmentation results of various approaches on the DFC2020 dataset. MURLF demonstrates exceptional visualization results, as highlighted by the red boxes. In the first group of experiments, the surface roughness of forests and grasslands is similar, posing a challenge for differentiation based solely on SAR images, while the two classes are distinguishable in optical images. The proposed MURLF effectively leverages the information characteristics of optical and SAR modalities, enabling accurate segmentation between these two classes. In the second and third groups of experiments, MURLF outperforms other methods by displaying finer details and achieving higher continuity in detecting water and grassland. In the fourth set of experiments, MURLF successfully detects the water within the green boxes, which lack annotations in the ground truth (GT). This demonstrates that the fine-tuned model avoids overfitting on the dataset and maintains robustness while delivering exceptional performance.

The segmentation results of various approaches on the WHU-OPT-SAR dataset are shown in Fig. 6. In the first group of experiments, distinguishing between forests and water is challenging solely based on optical images due to their similarity. However, the smooth surface of the water causes electromagnetic waves to undergo specular reflection, resulting in reduced brightness in SAR images. Conversely, the rough surface of forest regions leads to diffuse reflection, resulting in a brighter appearance in SAR images. Comparison methods tend to overly emphasize features in optical images, resulting in insufficient ability to distinguish between water and forests. The proposed MURLF effectively leverages information from both optical and SAR modalities to achieve accurate segmentation between water and forests. From the second, third, and fifth groups of experiments, it is evident that MURLF demonstrates higher continuity in road segmentation.

However, when characteristics in both the modalities are unclear, the continuous segmentation of narrow water surfaces remains challenging for all the approaches, as the purple boxes in the fifth group of experiments demonstrate. Addressing this challenge may require enhancing the model's global understanding of the scene, which could involve incorporating additional modal inputs or designing a more efficient downstream segmentation task head.

D. Single-Modality Scene Classification

Table VI illustrates the performance of various SSL-based methods in single-modality scene classification, where "MAE-OPT" indicates pretraining with the optical modality of the entire SSL4EO-S12 dataset, while "MAE-SAR" indicates pretraining with only the SAR images. We fine-tune using 10% of the BigEarthNet-MM training set. It is evident that using both the modalities during pretraining enhances accuracy in single-modality scene classification. Our analysis suggests that using both the modalities for pretraining enables the model to capture informational interactions between them, thereby improving performance in downstream tasks. The proposed MURLF improves mAP_{micro} by 2.6% for optical images and

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TABLE IV
COMPARATIVE RESULTS OF DIFFERENT METHODS ON THE DFC2020 DATASET FOR LAND COVER SEGMENTATION

Method	C1	C2	C3	C4	C5	C6	C7	C8	OA	mFscore	mIOU
Deeplabv3+	92.59	63.93	77.46	41.66	83.99	86.02	53.71	99.31	86.47	76.23	64.24
MCANet	92.89	58.16	78.03	45.16	81.17	86.01	60.29	99.09	85.93	75.71	63.57
CMX	93.89	**65.84**	78.90	47.57	86.35	87.10	54.05	99.46	87.48	77.83	66.53
Tokenfusion	93.57	68.58	76.71	**46.53**	85.95	87.13	64.02	99.37	87.33	77.57	66.46
ViT+UperNet	92.97	61.67	71.75	32.63	79.76	84.40	45.76	99.26	84.47	72.81	60.39
Swin+UperNet	91.77	55.37	**80.18**	44.80	85.25	86.07	64.96	99.40	86.65	76.76	64.82
DINO	93.52	63.56	76.02	45.30	85.73	87.05	56.37	**99.47**	87.16	77.62	65.76
MoCov3	93.31	63.06	76.72	45.46	85.21	86.26	57.11	99.45	87.09	77.26	65.59
MAE	93.21	61.45	77.86	41.34	85.22	85.98	57.54	99.39	86.93	76.93	65.22
S2FL	90.00	56.28	67.37	25.77	82.98	82.40	51.09	98.97	83.27	70.84	58.32
MultiMAE	**94.79**	62.20	76.33	46.13	86.70	85.96	54.40	99.46	87.26	77.83	66.27
MURLF	93.71	65.48	78.87	46.38	**86.80**	**87.76**	**67.22**	99.44	**88.13**	**79.40**	**68.05**
MURLF (ViT-B)	93.66	72.08	79.34	45.34	86.18	87.24	71.28	99.41	88.41	80.09	68.77

C1: Forest. C2: Shrubland. C3: Grassland. C4: Wetlands. C5: Croplands. C6: Urban-Built-up. C7: Barren. C8: Water.

TABLE V
COMPARATIVE RESULTS OF DIFFERENT METHODS ON THE WHU-OPT-SAR DATASET FOR LAND COVER SEGMENTATION

Method	C1	C2	C3	C4	C5	C6	C7	OA	mFscore	mIOU
Deeplabv3+	81.31	71.10	59.30	77.03	88.83	20.93	12.61	80.05	60.93	47.89
MCANet	84.19	64.10	66.49	80.33	90.29	56.88	37.81	82.94	70.14	56.12
CMX	83.96	**79.77**	64.25	80.64	89.95	54.46	31.32	83.23	70.70	56.88
Tokenfusion	84.86	76.34	64.35	80.29	90.00	53.13	33.52	83.49	71.13	57.28
ViT+UperNet	83.49	71.44	58.15	78.59	89.38	42.27	24.49	81.59	66.73	52.77
Swin+UperNet	83.25	75.04	61.02	78.74	89.80	44.34	19.81	81.96	67.14	53.36
DINO	84.03	75.06	61.79	80.97	90.13	50.19	29.89	82.94	69.89	56.02
MoCov3	84.25	73.04	61.95	79.90	89.45	50.26	29.33	82.62	69.52	55.55
MAE	84.13	75.65	62.81	80.55	90.19	50.90	31.18	83.06	70.34	56.46
S2FL	83.24	73.20	60.70	80.77	89.92	49.38	29.30	82.40	68.99	54.98
MultiMAE	84.00	75.59	63.46	81.12	90.36	53.89	30.82	83.22	70.64	56.81
MURLF	**85.09**	75.45	**67.64**	**82.73**	**90.60**	**60.50**	**36.64**	**84.31**	**72.88**	**59.24**
MURLF (ViT-B)	87.48	79.13	66.43	83.11	88.88	58.76	43.22	84.74	73.69	60.88

C1: Farmland. C2: City. C3: Village. C4: Water. C5: Forest. C6: Road. C7: Others.

5.65% for SAR images compared with using single-modality inputs for pretraining, leading to optimal performance.

E. Ablation Study

In this section, we comprehensively study the factors influencing the performance of the model. All the experiments proceeded with pretraining on the SSL4EO-S12 dataset, followed by fine-tuning on the BigEarthNet-MM dataset.

1) PSFE: Table VII presents an analysis of the influence of PSFE, revealing a substantial improvement upon integration of the PSFE module. Specifically, there is a notable improvement of 1.08% in mAP_{micro} compared with the baseline model. PSFE uses distinct structures to accommodate diverse modal inputs, facilitating the comprehensive extraction of information from each modality. This strategy stands in contrast to the conventional practice of using uniform feature extraction methods across modalities, thereby yielding superior model performance.

2) Depth of DFEE: Table VIII offers the effect of distinctive encoder depth on model performance. It is observed that the model exhibits relatively poorer performance when the distinctive encoder depth is set to 0. This can be ascribed to the substantial disparateness in features between optical and SAR modalities, suggesting that relying solely on a common feature encoder for both the modalities may result in suboptimal feature extraction. As the depth of the distinctive encoders

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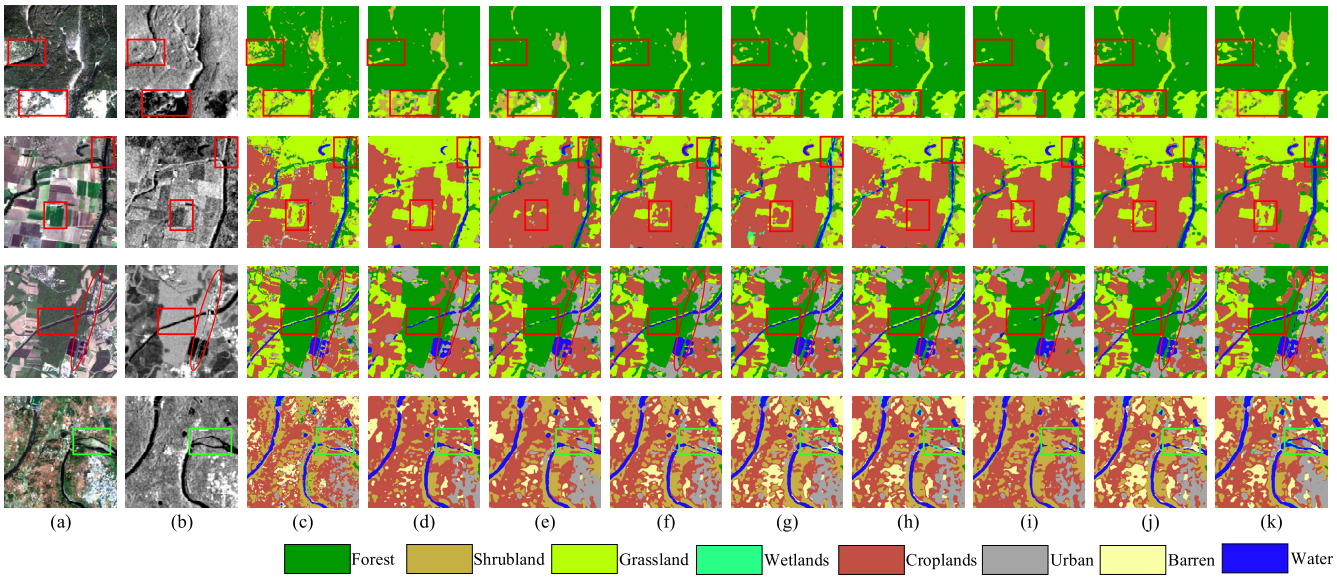


Fig. 5. Visualation results of MURLF and comparison methods on the DFC2020 dataset. (a) Optical images. (b) SAR images. (c) GT. (d) Deeplabv3+. (e) MCANet. (f) DINO. (g) MoCov3. (h) MAE. (i) S2FL. (j) MultiMAE. (k) MURLF.

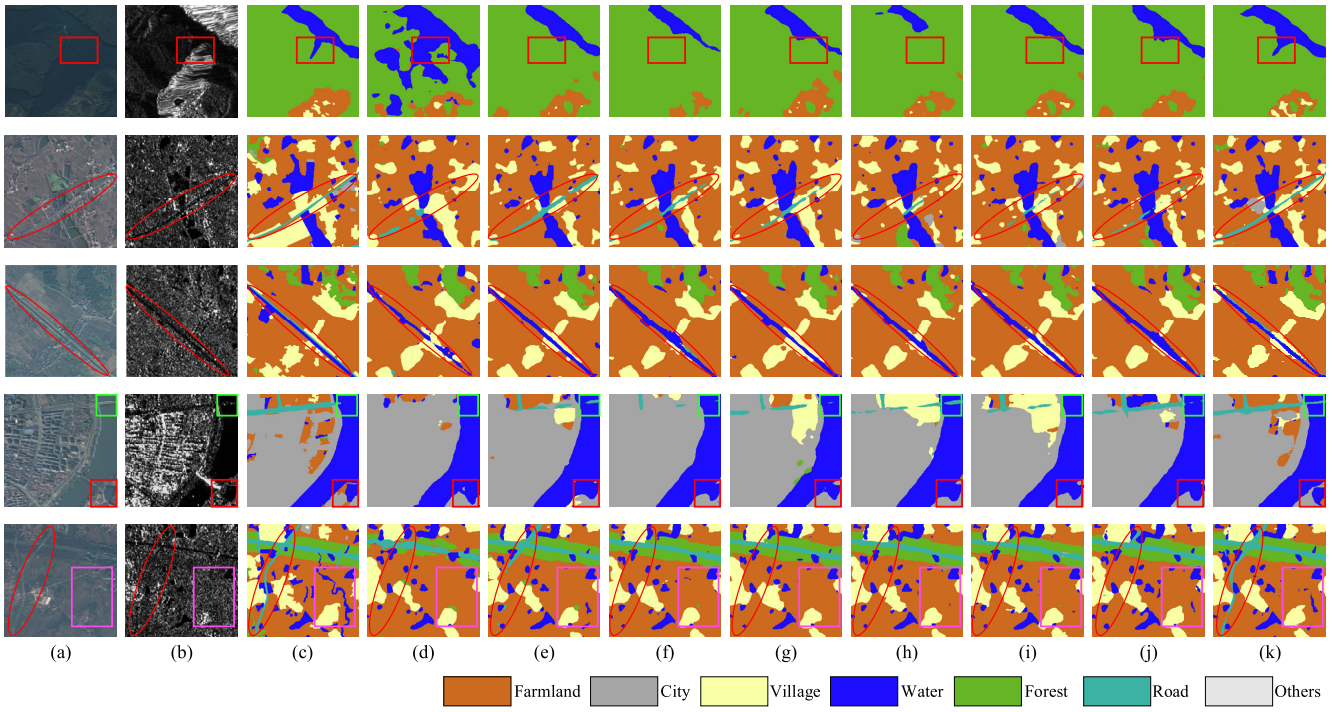


Fig. 6. Visualation results of MURLF and comparison methods on the WHU-OPT-SAR dataset. (a) Optical images. (b) SAR images. (c) GT. (d) Deeplabv3+. (e) MCANet. (f) DINO. (g) MoCov3. (h) MAE. (i) S2FL. (j) MultiMAE. (k) MURLF.

increases, the model's performance improves. Notably, even with the distinctive encoder set to a depth of 1, there is a noteworthy enhancement in performance, underscoring the efficacy of the model's structural design. We observe a slight performance decline when the DFEE depth is set to 4. This might be because a deeper DFEE design emphasizes features too heavily within DFEE, thereby reducing the extraction of effective common features by CFEE and potentially leading to overfitting.

3) Masking Strategies: We analyze the impacts of various masking strategies and mask ratios, as depicted in Tables IX and X. It is evident that independent masking proves to be more effective in enhancing performance across various modalities. This strategy efficiently uses both single-modal and cross-modal information, facilitating interaction between modalities. Furthermore, the model performs optimally when the mask ratio is 75%.

F. Visualization Analysis

Extensive feature visualization experiments are conducted to analyze the reasons for the superior performance exhibited by MURLF. We use the *t*-stochastic neighbor embedding (t-SNE)

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TABLE VI
FINE-TING RESULTS OF DIFFERENT METHODS FOR SINGLE-MODALITY SCENE CLASSIFICATION

		MAE-OPT	MAE-SAR	MAE	S2FL	MultiMAE	MURLF
OPT	mAP_{micro}	84.90		85.21	85.18	86.17	**87.51**
	mAP_{sample}	88.72		88.97	88.61	89.23	**90.97**
SAR	mAP_{micro}		74.63	76.30	75.81	78.21	**80.28**
	mAP_{sample}		81.30	82.41	81.50	83.69	**85.26**

TABLE VII
ABLATION ANALYSIS OF THE PSFE

	mAP_{micro}	mAP_{sample}
Baseline	90.53	93.08
Self-attention	90.69	93.33
PSFE	**91.61**	**93.97**

TABLE VIII
INFLUENCE OF DFEE DEPTH ON MODEL PERFORMANCE

Depth	MURLF	MURLF (ViT-B)
0	90.41	91.24
1	91.28	91.93
2	**91.61**	92.29
4	91.39	**92.32**

TABLE IX
INFLUENCE OF MASKING STRATEGY ON MODEL PERFORMANCE

Masking strategy	mAP_{micro}
Same position	90.80
Independent position	**91.61**

TABLE X
INFLUENCE OF MASK RATIO ON MODEL PERFORMANCE

Mask ratio	mAP_{micro}
60%	91.52
70%	91.55
75%	**91.61**
80%	91.49

algorithm to visualize the feature representations extracted by various MIM-based models on the BigEarthNet-MM dataset [82], as illustrated in Fig. 7. Specifically, we choose the single-label data in the test set for t-SNE. All the models are obtained from pretraining and have not undergone any fine-tuning processes. Our proposed MURLF exhibits notably apparent feature separation compared with the comparative methods. This enhancement showcases the capacity of the proposed MURLF to exploit more discriminative features.

Fig. 8 presents the heat map visualization of MURLF applied to the BigEarthNet-MM dataset. It can be seen that there are apparent differences in the activation regions between optical and SAR images within the same class, indicating MURLF's effective utilization of different features from both the modalities. Besides, even in the absence of pixel-level GT as a constraint, MURLF can still delineate the classes of different pixels with a high degree of accuracy. In addition, it is evident that the SAR modality exhibits a stronger emphasis on identifying regions characterized by distinctive scattering features, such as water and urban areas, while the optical modality excels at accurately differentiating between various types of vegetation. By making full use of the optical and SAR images, MURLF comprehensively understands the scene content.

V. EXPERIMENTS ON PAIRED RGB–NIR DATA

Considering the different reflection characteristics of RGB and near-infrared (NIR) data for various land cover types, the joint utilization of these two modalities can significantly enhance the model performance in various downstream tasks. Therefore, we use RGB–NIR paired data to further validate the effectiveness of our proposed MURLF.

A. Pretraining Setup

We extract the RGB (B4, B3, B2) and NIR (B8) bands from the SSL4EO-S12 dataset to create the SSL4EO-RGB-NIR dataset [32]. We conduct pretraining with the proposed MURLF and comparison methods for 100 epochs on this RGB–NIR dataset. It is worth noting that unless otherwise specified, all the parameter settings used in this section are consistent with those in Section IV.

B. Scene Classification

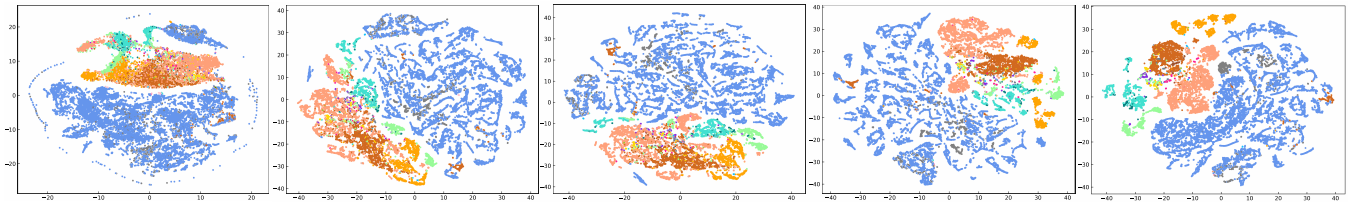
We use the RGB (B4, B3, B2) and NIR (B8) bands from the BigEarthNet-MM scene classification dataset to construct the BigEarthNet-RGB3NIR dataset [74]. We then perform fine-tuning for the scene classification task using MURLF and comparison methods. With the results presented in Table XI, MURLF leads across nearly all the metrics, showing an improvement of 1.4% in mAP_{micro} and 2.3% in mAP_{macro} compared with the second-best method. This underscores the effectiveness of MURLF for scene classification tasks using RGB–NIR data.

Compared with the accuracy of MURLF using both optical and SAR data as shown in Table II, using only RGB and NIR information results in a reduction of 1.47% in mAP_{micro} and 2.42% in mAP_{macro}. This outcome is expected, as including more input information helps the model more precisely distinguish between visually similar classes, thereby enhancing accuracy in downstream tasks.

C. Land Cover Segmentation

We use the DFC2020 land cover segmentation dataset, selecting the RGB (B4, B3, B2) and NIR (B8) bands to form the DFC2020-RGB–NIR dataset [77]. Performance comparisons between the proposed MURLF and other SOTA

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(a) (b) (c) (d) (e)

Fig. 7. t-SNE visualization of features obtained by MIM-based models on the BigEarthNet-MM dataset. (a) PCA. (b) MAE. (c) S2FL. (d) MultiMAE. (e) MURLF.

TABLE XI

COMPARATIVE RESULTS OF DIFFERENT METHODS ON THE BIGEARTHNET-RGB–NIR DATASET FOR SCENE CLASSIFICATION

Method	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17	C18	C19	mAP_{micro}	mAP_{macro}	mAP_{sample}
ResNet50	84.7	55.3	92.6	70.2	85.7	77.3	73.7	89.5	86.5	95.2	90.9	65.0	73.8	73.3	73.0	64.4	57.2	88.2	99.4	87.10	78.74	89.33
ViT	80.4	48.8	92.3	68.9	85.6	76.5	73.3	86.2	86.1	95.0	90.3	63.5	74.1	73.7	70.8	66.4	51.4	89.5	99.3	86.78	77.49	88.56
ExViT	81.9	50.1	92.6	71.7	85.8	77.5	74.1	87.9	86.1	95.0	90.8	65.2	75.3	74.9	70.3	67.6	54.9	89.5	99.5	87.30	78.48	89.77
SCT	85.3	55.6	93.1	73.6	86.0	77.3	75.0	89.5	87.0	94.9	91.1	64.8	76.6	75.8	73.9	68.6	57.3	90.9	99.6	87.89	79.80	90.39
Swin	85.0	56.0	93.1	70.9	86.3	76.8	74.7	89.4	86.9	95.3	90.8	64.0	74.1	75.3	70.1	69.8	55.5	90.8	99.5	87.78	79.18	90.20
DINO	86.7	58.2	93.8	75.2	86.9	78.9	75.7	90.5	87.7	95.7	91.7	66.5	76.7	76.7	72.8	72.0	58.8	92.0	99.6	88.62	80.86	91.98
MoCov3	84.8	57.0	93.5	73.5	86.6	78.0	75.3	88.6	87.1	95.5	91.5	65.6	76.0	76.3	72.7	71.9	58.5	91.7	99.6	88.33	80.19	91.68
MAE	84.7	57.1	93.3	73.9	85.7	76.1	72.9	88.1	86.5	94.7	90.4	67.1	78.6	75.2	79.3	72.9	64.7	93.6	99.6	87.67	80.77	91.60
S2FL	85.6	56.5	93.4	73.7	86.5	78.3	75.6	90.0	87.4	95.5	91.3	65.5	74.2	75.8	72.2	70.8	58.7	91.2	99.5	88.24	80.11	91.23
MultiMAE	86.2	58.5	93.6	76.2	86.1	77.8	75.1	88.5	87.6	95.3	91.5	67.4	78.9	76.7	80.5	73.3	64.1	**93.6**	99.6	88.71	81.62	92.05
MURLF	**88.3**	**62.4**	**94.6**	**80.3**	**88.2**	**80.7**	**77.9**	**92.3**	**88.9**	**96.2**	**92.6**	**71.8**	**83.3**	**79.2**	**79.6**	**76.8**	**68.1**	93.3	**99.7**	**90.14**	**83.91**	**93.03**

C1: Urban fabric. C2: Industrial or commercial units. C3: Arable land. C4: Permanent crops. C5: Pastures. C6: Complex cultivation patterns. C7: natural vegetation. C8: Agro-forestry areas. C9: Broad-leaved forest. C10: Coniferous forest. C11: Mixed forest. C12: Natural grassland. C13: Moors, heathland and sclerophyllous vegetation. C14: Transitional woodland/shrub. C15: Beaches, dunes, sands. C16: Inland wetlands. C17: Coastal wetlands. C18: Inland waters. C19: Marine waters.

TABLE XII

COMPARATIVE RESULTS OF DIFFERENT METHODS ON THE DFC2020-RGB–NIR DATASET FOR LAND COVER SEGMENTATION

Method	C1	C2	C3	C4	C5	C6	C7	C8	OA	mFscore	mIOU
Deeplabv3+	89.16	60.16	68.63	29.64	80.63	81.76	41.08	98.56	82.60	70.24	57.43
MCANet	89.15	59.95	67.20	32.73	80.42	67.32	61.57	98.56	81.55	69.43	56.45
CMX	91.71	53.27	70.28	35.60	86.46	81.75	58.46	99.24	84.89	73.58	61.22
Tokenfusion	90.67	53.32	68.16	30.22	82.12	81.11	33.07	98.58	82.52	68.97	56.42
ViT+UperNet	88.74	52.61	68.01	22.72	79.27	77.50	39.02	98.01	80.93	67.19	54.44
Swin+UperNet	90.03	53.48	67.37	21.79	77.96	78.89	43.17	98.39	81.35	67.71	55.01
DINO	91.91	52.96	72.82	31.82	82.31	82.33	45.32	99.20	83.98	71.74	59.25
MoCov3	90.43	53.41	69.24	29.73	83.19	79.21	43.91	98.71	82.91	70.39	57.64
MAE	91.81	55.57	71.52	30.40	83.00	82.11	44.09	99.04	83.98	71.66	59.17
S2FL	89.78	52.52	66.21	27.93	80.28	78.15	48.41	98.27	81.72	68.90	56.03
MultiMAE	91.70	55.47	72.42	33.29	83.42	83.01	49.70	99.24	84.48	72.73	60.30
MURLF	**93.09**	**66.84**	**76.47**	**45.34**	**84.90**	**86.24**	**55.60**	**99.32**	**87.00**	**77.38**	**65.49**

C1: Forest. C2: Shrubland. C3: Grassland. C4: Wetlands. C5: Croplands. C6: Urban-Built-up. C7: Barren. C8: Water.

methods are presented in Table XII. MURLF demonstrates a 4.7% improvement in mFscore and a 4.3% improvement in mIOU compared with the second-best method, confirming its effectiveness for land cover segmentation tasks using RGB–NIR data.

Compared with the performance of MURLF in Table IV, using only RGB and NIR information in Table XII results in a 1.13% reduction in OA. MURLF exhibits the least performance decline compared with other methods. This reduction is attributable to MURLF's ability to effectively consider

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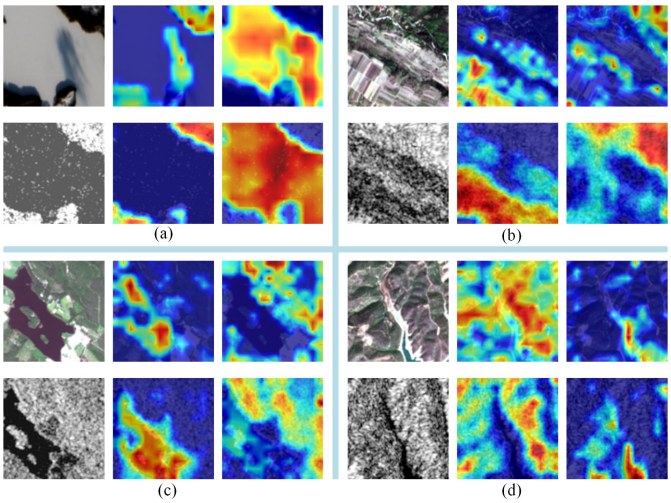


Fig. 8. Visualization of the heat map of the last layer on the BigEarthNet dataset. (a)–(d) Each represent four different scenes. In each scene, the first row shows the optical image and its corresponding heat maps for different classes. The second row displays the SAR image and its corresponding heat maps for different classes.

the unique characteristics of different modalities and their complementary relationships, thereby optimizing the use of available information.

VI. Conclusion

In this article, we propose MURLF, a novel MURLF based on MIM, to uncover potential associations among multimodal data. MURLF leverages the inherent consistency and complementarity among different modalities to address the challenges posed by significant feature differences on the encoder. The encoder comprises a CFEE and multiple lightweight DFEEs, enabling the model to maintain high accuracy without significantly increasing computational load, even when handling inputs from multiple modalities. In addition, MURLF uses same-modal and cross-modal information to recover masked tokens, promoting effective information interaction between modalities. The multistage PSFE is designed to comprehensively consider the characteristics of data from various sensors, fully retaining effective information from different modalities. Through the integration of the multistage PSFE with the ViT backbone, MURLF extracts multimodal hierarchical representations for downstream tasks. Notably, MURLF is not restricted to multimodal inputs but also supports single-modal inputs, significantly expanding its applicability. In the future, we plan to explore multimodal joint pretraining models with larger resolution differences, such as combining UAV and satellite images. We hope this work could bring inspiration for constructing multimodal foundation models using massive unlabeled data within the RS community.

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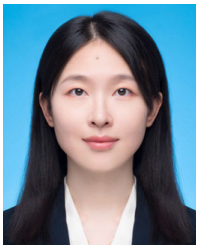
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